

A
Scope and data resources

scTHREAD integrates single-cell long-read transcriptome data from human and mouse across platforms and library designs.

34 datasets	453 run records	923,389 dataset-reported cells	Sequencing platforms 399 ONT runs 54 PacBio runs	Library design 155 barcode-resolved (droplet-based or other pooled-barcoded) 298 sample/file-resolved (plate-based or individually indexed)	Species Human Mouse
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B Unified processing workflow

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graph TD; A[Barcode-resolved<br/>(155 records)] --> D[Read processing<br/>and quality control]; B[Sample/file-resolved<br/>(298 records)] --> D; D --> E[Alignment<br/>and BAM organization]; E --> F[IsoQuant<br/>Transcript quantification<br/>and model construction]; F --> G[453 IsoQuant core outputs:<br/>✓ gene counts<br/>✓ transcript counts<br/>✓ read assignments<br/>✓ transcript models];
```

Barcode-resolved (155 records)

Sample/file-resolved (298 records)

Read processing and quality control

Alignment and BAM organization

IsoQuant
Transcript quantification and model construction

453 IsoQuant core outputs:

- ✓ gene counts
- ✓ transcript counts
- ✓ read assignments
- ✓ transcript models

C **scTHREAD evidence landscape**

UMAP of single-cell transcriptomes (illustrative)

The figure is a UMAP plot showing the relationship between UMAP 1 (x-axis) and UMAP 2 (y-axis). The plot displays several distinct clusters of points, each representing a different cell type. The clusters are color-coded and labeled: Immune cells (blue), Neuronal cells (orange), Epithelial cells (teal), Stromal cells (purple), Endothelial cells (red), and Muscle cells (dark blue). A legend on the right side of the plot, titled 'Major cell types (illustrative)', lists these cell types with corresponding colored circles. The clusters are distributed across the plot, with Immune cells in the upper left, Neuronal cells in the upper center, Epithelial cells in the upper right, Stromal cells in the lower right, Endothelial cells in the lower center, and Muscle cells in the lower left.

Major cell types (illustrative)

- Immune cells
- Neuronal cells
- Epithelial cells
- Stromal cells
- Endothelial cells
- Muscle cells

D

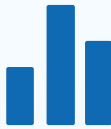
Data access



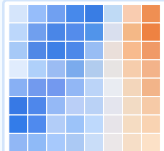
Search & Browse



datasets · genes · isoforms · cell types · regions



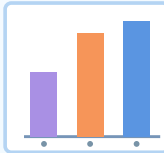
Cell Map & Expression



multi-isoform view · normalized heatmap
shared scales · effect cards · CSV/JSON export



Compare across studies



isoform usage · APA · junction · ASE
shared scales · effect cards · CSV/JSON export



Novel models & evidence



NIC/NNC · junction browser · provenance
shared scales · effect cards · CSV/JSON export



Web · API · TSV · JSON

E

Evidence layers and availability
Downstream evidence-layer availability reported independently.

Isoform evidence

- Transcript models
- Isoform expression
- Alternative splicing

Poly(A) evidence

- Polyadenylation sites
- 3' end evidence
- Isoform-level poly(A)

Junction evidence

- Splice junctions
- Junction support
- Novel junctions

Allelic evidence

- Allele-specific expression
- Heterozygous sites
- Isoform-level allelic support

Isoform evidence



- Transcript models
- Isoform expression
- Alternative splicing

Poly(A) evidence

 AAA... *

- Polyadenylation sites
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Junction evidence



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A

G

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